

Steps and requirements.

You need to follow the steps and document the output for MP distribution and Shrinkage estimator from yahoo finance page by merging the

Code:

1. Go to tests
2. Open—> main-Copy1.ipynb
3. Open —> dcmppln/denoiser-Copy1.py
 - a. Code for using MP distribution -
from dcmppln.utils.correlation_rmt import split_covariance_matrices
 - b. Code for using shrinkage estimator-
from dcmppln.utils.correlation_rmt_Copy1 import split_covariance_matrices
 - c. Code for using Bayesian Principal Component Analysis (PCA)
from dcmppln.utils.correlation_rmt_Copy2 import split_covariance_matrices
 - d. Code for using Permutation-Thresholded Sparse Covariance Decomposition with Shrinkage (PTSCDS)
from dcmppln.utils.correlation_rmt_Copy3 import split_covariance_matrices
 - e. Code for using Permutation-Thresholded Sparse Covariance Decomposition with Shrinkage (PTSCDS)
from dcmppln.utils.correlation_rmt_Copy3 import split_covariance_matrices
4. To see what data is being used—>tests/input_data.py, here you can check what data is in use. And you have to replace the data in this python file to see results in step-2.

How will be the workflow:(Make sure to save every part before you run and document the result)

After opening all of the above files—>

Step1 Integrates the yahoo finance code as per your understanding, to fetch data directly from NASDAQ, SENSEX etc.

Step2 keeps “3a.”, as it is in “3.” and run “2.”and document the result .(Make sure to save every part before you run)

Step3 Edit “3a.” with “3b/c/d/e.”. Document the result you get.

And repeat this for 8-10markets and send me the results with proper explanation and your observation.

Step-by-Step Guide for MP Distribution & Shrinkage Estimator Using Yahoo Finance Data

This guide will help you fetch financial data, apply MP distribution and shrinkage estimators, and analyze results for multiple markets. Follow the steps carefully and document every outcome.

Step 1: Open Required Files

Before you begin, navigate to the necessary files in your project directory:

1. **Go to `tests/` folder**
 - a. Open `main-Copy1.ipynb` (Jupyter Notebook)
 - b. Open `dcmppln/denoiser-Copy1.py`
 2. **Ensure the following import statements are correct:**
 - a. **For MP distribution:**

```
from dcmppln.utils.correlation_rmt import  
split_covariance_matrices
```
 - b. **For Shrinkage Estimator:**

```
from dcmppln.utils.correlation_rmt_Copy1 import  
split_covariance_matrices
```
 3. **Check Input Data Source:**
 - a. Open `tests/input_data.py`
 - b. This file contains market data; you will replace this data to analyze different markets.
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Step 2: Fetch Market Data from Yahoo Finance

Now, integrate a Yahoo Finance API call to fetch data directly from various stock markets (NASDAQ, SENSEX, etc.).

Step 3: Run the MP Distribution & Shrinkage Estimator

Now, execute the main steps in sequence and **document the output at each stage**.

Execution Workflow:

1. **Keep "2a." in place** inside the code.
 2. **Run Step "1a." and document the results.**
 - Before executing, save all modified files.
 - Run the script, observe the output, and note any anomalies.
 - Record observations, including statistical insights.
 3. **Modify "2a." → "2b."** in your code.
 4. **Re-run the script and document the results.**
 - Compare with previous results.
 - Highlight differences between "2a." and "2b.".
 5. **Repeat for 8-10 different markets.**
 - Modify `fetch_market_data()` with different tickers.
 - Run and document results for each market.
 - Compare trends across multiple markets.
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Final Submission

What You Need to Provide:

- Code modifications
- Results
- Comparison of "2a." vs. "2b." outputs
- Market-wise insights (for 8-10 markets)

Link for the screen

recording=<https://drive.google.com/file/d/1TsndLalHpTLah78n6vCc8VuvexLmA0uS/view?usp=sharing>

#step-change the /dcmppln/utlils/correlation_rmt----> JP_VG/dcmppln/denoiner---->
JP_VG/dcmppln/pipeline---->example/main

